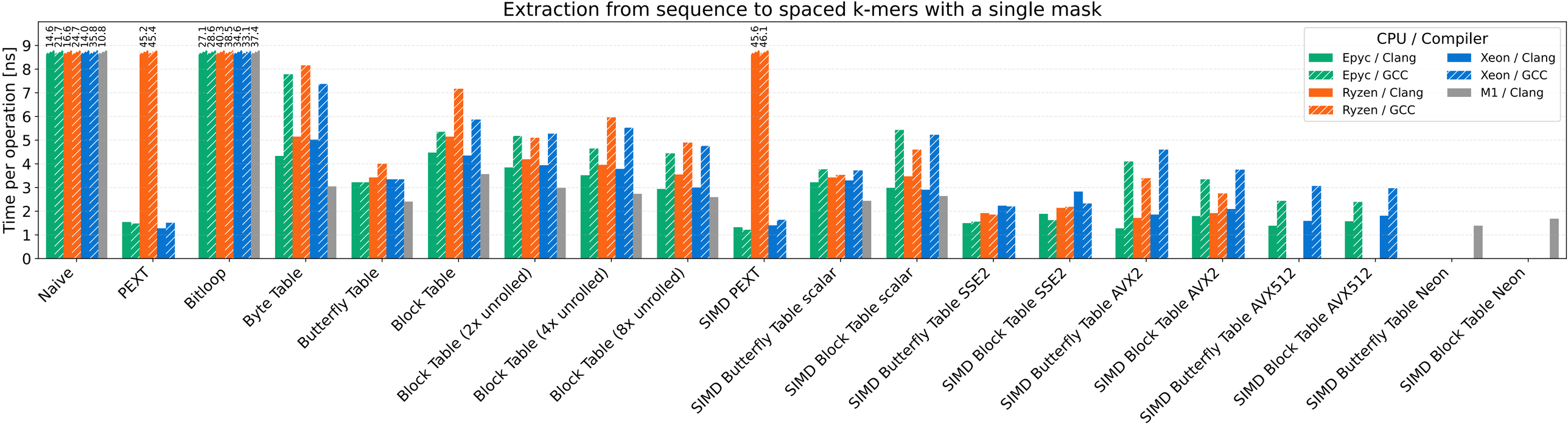
Extraction from sequence to spaced k-mers with a single mask
Time per operation [ns]
0
1
2
3
4
5
6
7
8
9
14.6
21.7
16.6
24.7
14.0
35.8
10.8
45.2
45.4
27.1
28.6
40.3
38.5
34.6
33.1
37.4
45.6
46.1
CPU / Compiler
Epyc / Clang
Epyc / GCC
Ryzen / Clang
Ryzen / GCC
Xeon / Clang
Xeon / GCC
M1 / Clang
Naive
PEXT
Bitloop
Byte Table
Butterfly Table
Block Table
Block Table (2x unrolled)
Block Table (4x unrolled)
Block Table (8x unrolled)
SIMD PEXT
SIMD Butterfly Table scalar
SIMD Block Table scalar
SIMD Butterfly Table SSE2
SIMD Block Table SSE2
SIMD Butterfly Table AVX2
SIMD Block Table AVX2
SIMD Butterfly Table AVX512
SIMD Block Table AVX512
SIMD Butterfly Table Neon
SIMD Block Table Neon